Chapter 10

# Pineapple

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## 10.1 Introduction

The global volume of food waste has been recently estimated at 1.6 billion tonnes, resulting in 3.3 billion tonnes of $CO_2$ released into the atmosphere per year (FAOSTAT, 2015). The large majority of this waste, about 42%, has been generated by households, 38% due to food processing, 14% comes from food catering/services sector, and 5% from retail/wholesale. Vegetable and fruit production followed by the dairy, poultry, fishery, and winery industries are the main food sectors generating large volumes of waste (Baiano, 2014). These proportions are summarized in Table 10.1.

Normally, the waste from the food industry is used as animal feed, sent for composting or base for fertilizers, or mostly brought to landfills. Unfortunately a significant portion is often unscrupulously disposed of since the cost of disposal may be significant due to high transportation cost and limited availability of landfills (Rani and Nand, 2004), which in turn results in environmental issues. Several food waste handling issues require further consideration, such as (Laufenberg et al., 2003):

- prevention of environmental pollution;
- conservation of energy and new materials; and
- new methods and policies on waste handling.

The conversion of food waste into useful by-products of higher value, or even as raw materials for other industries as well as food or feed/fodder after biological treatment, has appeared to be a potential avenue for turning waste into a profitable material. Based on these points, bioconversions appear as an attractive solution due to the fact that these residual matters can be potentially converted into useful products. Numerous reports have emerged more recently on waste utilization, especially waste obtained from fruits and vegetables, for further industrial processes including fermentation, bioactive components extraction, etc. (Upadhyay et al., 2010).

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**TABLE 10.1** Estimate of waste generated in the food industry (Baiano, 2014).

| Industrial sector | Waste (%) |
|---|---|
| Production, processing, and preservation of meat and meat products | 8 |
| Production and preservation of fish and fish products | 0.4 |
| Production and preservation of fruits and vegetables | 14.8 |
| Manufacturing of vegetable and animal oils and fats | 3.9 |
| Dairy products and ice cream industry | 21.3 |
| Production of grain and starch products | 12.9 |
| Manufacture of other food products | 12.7 |
| Drinks industry | 26 |
| Total | 100 |

The waste from the fruit and vegetable industry consists of fruit and vegetable trimmings, peelings, stems, seeds, shells, cereal residues (such as bran), starch, sugar and juice extraction, and off-spec or damaged products (Baiano, 2014). In addition, some other high-value components can be extracted such as proteins, polysaccharides, fibers, flavor compounds, and phytochemical ingredients. These potentially marketable components can be used in various applications, such as pharmaceutical, cosmetic, food, and nonfood areas. The recovery of these high-value components is an elegant way to reuse the waste streams, as well as being economically interesting (Laufenberg et al., 2003). Among all the fruits and vegetables produced for this industry, pineapple is one of the world's major crops with a high production each year.

## 10.2 Pineapple waste utilization

Pineapple (*Ananas comosus* L.) is an edible member of Bromeliaceae family, and one of the important fruits in the world. According to FAO online database, the world pineapple production in 2014 was 24,778,262 tonnes with a yield of 242,281 hectogram per hectare (Hg/ha) and the number is expected to further increase by year (FAOSTAT, 2015). The fruit is consumed fresh or commercially processed into canned fruits, juice, concentrate, and jam. A steady rise in pineapple production, especially canning, has generated a large amount of waste due to selection and the elimination of unwanted parts (Tanaka et al., 1999), in which 75% of the unwanted parts are in the form of pineapple peel, core, and crown end (Fig. 10.1, Roda et al., 2016a,b). In addition, up to 55% of the generated waste is usually created due to poor fruit handling and exposure to adverse temperature conditions during storage and transport (Nunes et al., 2009). Fortunately the pineapple waste contains many reusable, highly valuable substances that require the application of novel scientific and technological extraction methods. The chemical properties of the different pineapple waste parts are summarized in Table 10.2 (Upadhyay et al., 2010). As indicated, many high-value components such as cellulose (insoluble fiber), protein mainly bromelain, pectin (soluble fiber), and simple sugars are available in different parts and forms of the pineapple waste. Hence numerous efforts on waste utilization for further industrial processes in various fields have been performed, some which are summarized in Table 10.3 and one of the approaches is illustrated in Fig. 10.2. Based on this information, the residues from the pineapple industry can be applied in various fields due to the multifunctional nature of these components. Furthermore, the waste has attractive potential for large-scale utilization due to being readily available, practically free, and renewable. In the following sections some aspects of extraction of valuable components will be provided. While the extraction methods may be numerous, only those that may be scaled up to a commercial level will be covered.

## 10.3 Protein utilization from pineapple waste—bromelain enzyme

Bromelain is a crude extract from pineapple (*A. comosus* L.) that contains a mixture of different proteases as well as phosphatases, glucosidases, peroxidases, cellulases, glycoproteins, and carbohydrates (Feijoo-Siota and Villa, 2010). Different types of protease in the bromelain mixture are summarized in Table 10.4 (Barrett et al., 2013).

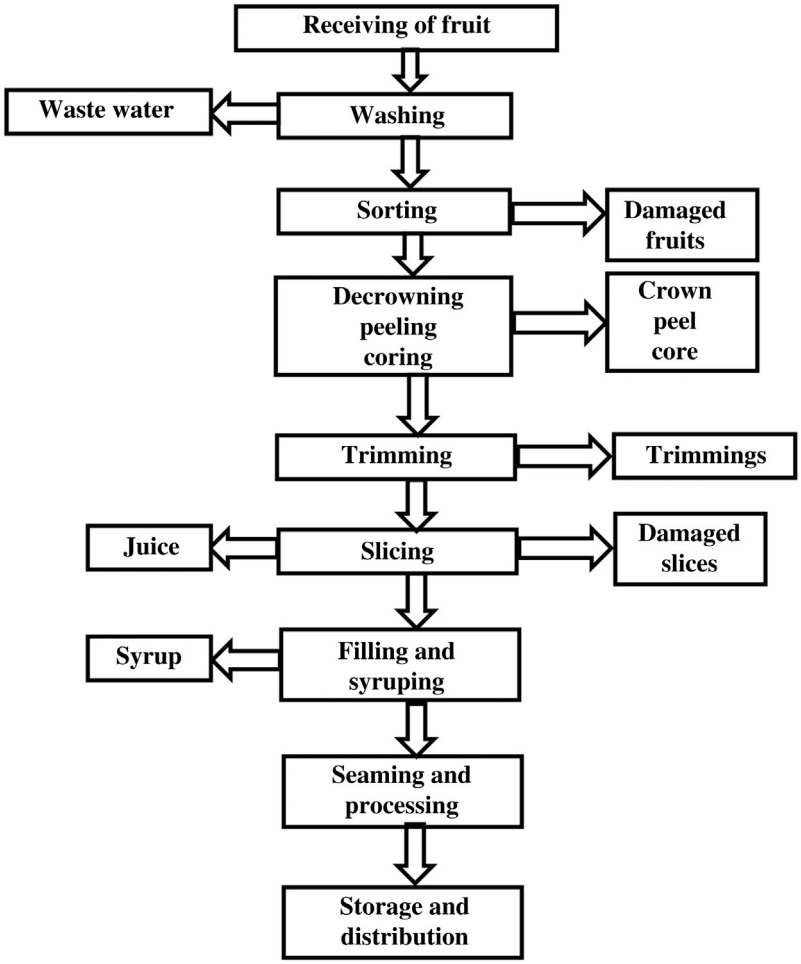



**FIGURE 10.1** Generation of waste during canning of pineapple.

**TABLE 10.2** Chemical properties of pineapple waste.

| Parameter | Peel[a] (ensiled) | Peel[a] (fresh) | Peel[a] (dry) | Whole[b] | Skin[b] | Crown[b] | Pulp[b] | Whole[c] | Peel[d] |
|---|---|---|---|---|---|---|---|---|---|
| Moisture (%) | 72.49 | 71.07 | 27.43 | – | – | – | – | 87.5 | 92.2 |
| Total solid (%) | 27.51 | 29.03 | 72.57 | – | – | – | – | 12.5 | 7.8 |
| Volatile solid (%) | 87.12 | 96.12 | 95.9 | – | – | – | – | – | 89.4 |
| pH | 4 | 4.7 | 4.7 | – | – | – | – | – | – |
| Ash (%) | 12.88 | 3.88 | 4.1 | 0.7 | 0.6 | 0.4 | 0.2 | 4.05 | 10.6 |
| *As % dry basis* | | | | | | | | | |
| Cellulose | 9 | 11.2 | 12 | 19.4 | 14 | 29.6 | 14.3 | – | 19.8 |
| Hemicellulose | 4.7 | 7 | 6.5 | 22.4 | 20.2 | 23.2 | 22.1 | – | 11.7 |
| Pectin | 5.1 | 6.7 | 7.1 | – | – | – | – | – | – |

(*Continued*)

**TABLE 10.2** (Continued)

| Parameter | Peel[a] (ensiled) | Peel[a] (fresh) | Peel[a] (dry) | Whole[b] | Skin[b] | Crown[b] | Pulp[b] | Whole[c] | Peel[d] |
|---|---|---|---|---|---|---|---|---|---|
| Ether soluble solid | 4 | 6.1 | 6.7 | – | – | – | – | – | – |
| Protein | 0.91 | 3.13 | 3.3 | 4.4 | 4.1 | 4.2 | 4.6 | 5.18 | – |
| Reducing sugar | 5 | 25.8 | 27.8 | 6.5 | – | – | – | 20.93 | – |
| Nonreducing sugar | 1.7 | 5.7 | 4.9 | 5.2 | – | – | – | – | – |
| Total sugar | – | – | – | 11.7 | – | – | – | – | – |
| Lignin | 9 | 11.52 | 11 | 4.7 | 1.5 | 4.5 | 2.3 | – | – |

[a] *Rani and Nand (2004)*.
[b] *Ban-Koffi and Han (1990)*.
[c] *Abdullah and Mat (2008)*.
[d] *Bardiya et al. (1996)*.

**TABLE 10.3 Pineapple waste utilization in numerous applications (Nor, 2017).**

| Application | Description | References |
|---|---|---|
| Bromelain | A protease enzyme extracted from pineapple. This enzyme has various applications in the pharmaceutical, medical, food, cosmetic, and textile industries | Murachi (1970), Murachi et al. (1964), Takahashi and Murachi (1976), Yamada et al. (1976) |
| Ethanol production | Pineapple waste is fermented by microorganisms and catalysts to produce ethanol | Nigam (1999, 2000), Tanaka et al. (1999) |
| Phenolic antioxidants | High phenolic compounds that are useful for health such as myricetin, salicylic acid, tannic acid, *trans*-cinnamic acid, *p*-coumaric acid, syringic acid, and ferulic acid have been identified in pineapple waste. These compounds are extracted from the waste by aqueous methods by using solvents such as ethanol and methanol | Larrauri et al. (1997), Upadhyay et al. (2012) |
| Organic acid | Pineapple waste is fermented to produce organic acids such as citric, lactic, and ferulic acids which can be commercially used in food and nonfood industries | Idris and Suzana (2006), Kumar et al. (2003), Tilay et al. (2008) |
| Energy and carbon source | Biomethanation of pineapple waste by anaerobic digestion and composting produces methane gas, while stabilizing effluent with a neutral pH and odorless property. Besides, the waste was utilized as a carbon source to produce hydrogen gas from municipal sewage sludge | Bardiya et al. (1996), Wang et al. (2006) |
| Antidyeing agent | Pineapple waste is used as a low-cost absorbent to remove basic dyes from polluting the wastewater in textile industry | Hameed et al. (2009), Weng et al. (2009) |
| Fiber | Fiber from pineapple leaves is utilized in the making of course textile, threads, and polymeric composites | Arib et al. (2006), Devi et al. (1998), Tran (2006) |
| Removal of heavy metal | Pineapple waste is used as an effective bioabsorbent to remove toxic metals like mercury, lead, cadmium, copper, zinc, and nickel from contaminated sewage sludge | Dacera and Babe (2008), Senthilkumaar et al. (2000) |
| Animal feed | Dried and ensiled pineapple wastes are utilized as feed for ruminants | Sruamisri (2007), Tran (2006) |

Of the different proteases available in a bromelain mixture, commercial bromelain is usually produced from the stem part (stem bromelain), although other pineapple parts contain a certain amount of bromelain as well. Ketnawa et al. (2012) reported the bromelain activity and protein content from different pineapple parts of waste such as peel, core, stem, and crown in *Nang Lae* and *Phu Lae* cultivars. They found that the extract from each waste part exhibited

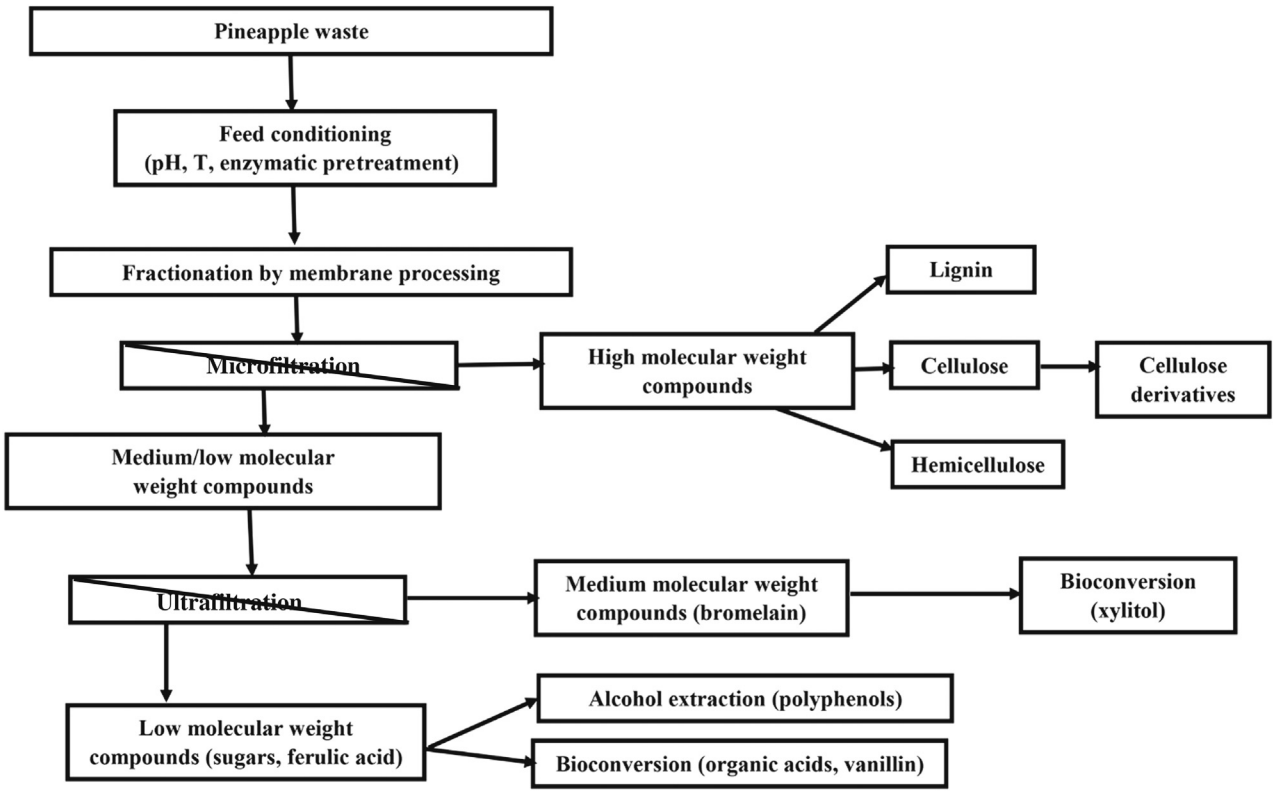



**FIGURE 10.2** Illustrative application of membrane processing in the valorization of pineapple waste into highly valuable compounds.

**TABLE 10.4** Different proteases in bromelain mixture (Barrett et al., 2013).

| Name | Source | EC number | Molecular weight | Isoelectric point |
|---|---|---|---|---|
| Stem bromelain | Pineapple stem | EC 3.4.22.32 | 24.5 | 9.55 |
| Fruit bromelain | Pineapple fruit | EC 3.4.22.33[a] | 25 | 4.6 |
| Pinguinain | Pineapple fruit | EC 3.4.22.33[a] | 26 | 6.5 |
| Ananain | Pineapple stem | EC 3.4.22.31 | 23.5 | >10 |
| Comosain | Pineapple stem | Not available | 24.5 | >10 |

[a]*Fruit bromelain and pinguinain are classified under the same EC number.*

different proteolytic activity and protein content, with the highest value obtained from the crown part, while the extract from the stem had the lowest value in both cultivars. These findings give a promising opportunity to extract such enzymes sustainably from waste sources.

Bromelain has been widely used in pharmaceutical and medical, food, detergent, cosmetic, and textile industries. Some of the major industrial applications of bromelain are summarized in Table 10.5, and have been reviewed by Arshad et al. (2014), Bhattacharyya (2008), Novaes et al. (2016), and Pavan et al. (2012). In the food industry, bromelain is mainly used for meat tenderization since this enzyme can effectively hydrolyze meat myofibril proteins, and is commercially marketed as a meat tenderizer under brands such as McCormick and Knorr (Arshad et al., 2014). It is also used to inhibit fruit browning by preventing the oxidation of phenol to quinone (Ramalingam et al., 2012), to enhance protein stability of beer and wine in alcohol production (Benucci et al., 2011), to improve dough relaxation, and to enhance stability and prevent dough shrinkage in the baking industry (Kong et al., 2007). The utilization of bromelain in fish protein hydrolysis has generated fish protein hydrolysate that possesses nutritional functions and

**TABLE 10.5 Industrial applications of bromelain (Nor, 2017).**

| Industry | Applications |
|---|---|
| Food | Meat tenderization |
| | Fruit antibrowning agent |
| | Alcohol production |
| | Dough improver |
| | Fish protein hydrolysate production |
| | Food supplement |
| | Animal food |
| Pharmaceutical and medical | Antiinflammatory agent |
| | Anticancer agent |
| | Digestive aid |
| | Potential as antibiotics |
| | Debridement cream |
| | Platelet aggregation inhibitor |
| | Arthritis preventer |
| | Antidiarrhea |
| Cosmetic | Wrinkle treatment |
| | Dry skin treatment |
| | Tooth whitening |
| | Reduce postinjection bruising |
| Textile | As cocoon softener in silk production |
| | Improve dyeing properties |
| Detergent | Detergent formulation |

antioxidant characteristics (Tanuja et al., 2012). Furthermore, bromelain application in animal feed can be used for protein degradation in the feed, increasing the protein inversion and availability, and subsequently decreasing the cost of the feed (Arshad et al., 2014; Tomankova and Kopecny, 1995).

The therapeutic application of bromelain has been extensively studied, and the antiinflammatory effects of bromelain has been reported since 1960s (Novaes et al., 2016). This is proven by inflammation pain inhibition in rats in a dose-dependent manner, with further reports of postoperative swelling and pain reduction (Bhattacharyya, 2008). Bromelain is an effective anticancer therapeutic agent, as a result of the systemic response to multiple cellular and molecular targets induced by this enzyme; however, the real underlying mechanisms are yet to be fully understood (Novaes et al., 2016). Also this enzyme has the potential to be used as an antibiotic since it can modify the permeability of organs and tissues to different drugs (Bhattacharyya, 2008). It is reported that bromelain can be used as digestive enzyme in intestinal disorders, suggesting its application as a digestive aid (Knill-Jones et al., 1970; Rathnavelu et al., 2016). Moreover bromelain is used in a debridement cream formulation to eliminate burn debris and accelerate the healing process, and can inhibit platelet aggregation especially for the patients affected with heart attack and stroke, reduce arthritis pathogenesis since it has analgesic properties that can influence pain mediators, and prevent diarrhea by interacting with intestinal secretory signaling pathways (Bhattacharyya, 2008; Pavan et al., 2012).

In addition to food and medical applications, bromelain is used in the cosmetic industry to treat various skin conditions such as low skin firmness, wrinkles, and dry skin (Ozlen, 1995). It is also used as a stain remover in teeth cleaning/whitening (Chakravarthy and Acharya, 2012). In the textile industry it is used to improve the dyeing properties of protein fibers such as wool and silk (Koh et al., 2006) and assist in the cocoon-softening process in silk production (Singh et al., 2003).

## 10.4 Bromelain extraction strategies

Due to various applications, commercial bromelain is very expensive, costing up to US$2400/kg (Spir et al., 2015) primarily due to its high production cost. Thus the development of effective concentration and purification techniques by using waste from the pineapple industry is a current concern in this area, with the aim to minimize the number of operations required to obtain the desired enzyme purity.

Various purification strategies used for bromelain have been reviewed extensively (Arshad et al., 2014; Bala et al., 2012; Manzoor et al., 2016; Nadzirah et al., 2013). They normally follow the typical protein purification strategies which can be performed with different separation techniques (Table 10.6). These include (1) chromatography, (2) precipitation, (3) aqueous two-phase system (ATPS), (4) reverse micellar system (RMS), (5) membrane filtration, or (6) the integration of any of these techniques (Table 10.6; Arshad et al., 2014). For each purification approach, the method's efficiency is evaluated in terms of recovery yield and purity increment of the enzyme. In terms of high enzyme purity and selectivity, chromatography appears to be the best option in comparison to other purification techniques (Nadzirah et al., 2013). Nevertheless, most of the bromelain commercial applications do not require high purity and the enzyme is normally produced as crude in high quantity for applications in high-volume productions (Illanes, 2008).

At the industrial level the selection of a suitable purification technique should be focused on the feasibility and practicability of its implementation in a large-scale operation. Since the purification cost can reach up to 80% of the total production cost, the development of low-cost and viable upscalable techniques should be considered (Spir et al., 2015). The advantages and disadvantages of each purification technique are listed in Table 10.7 (Arshad et al., 2014). Among all the mentioned purification techniques (Tables 10.6 and 10.7), chromatography is the least suitable for large industrial-scale application due to its small sample loading capacity that requires multiple purification steps, and thus it is only suitable for small-scale bromelain production.

The precipitation method can be easily scaled up; however, it requires the use of different chemicals which are not technologically green. Some restrictions of the ATPS method, including limited range of polarities between the coexisting phases, high cost due to employed polymers, and high viscosity, have limited its potential (Vicente et al., 2016). In addition, difficulties associated with recovering the protein of interest from a solvent final mixture in the RMS method present a serious obstacle for full application of this technology.

Membrane filtration technology is an attractive option since it can be easily scaled up with high-throughput, and it is a green technology due to no chemical requirement (Girard and Fukumoto, 2000; Hinkova et al., 2002). It involves minimal operation cost compared to other purification techniques, including the liquid−liquid extraction method (Lopes et al., 2012). This technique has been successfully applied in many protein separation areas that require high purity and efficient operation (Lipnizki, 2010; van Reis and Zydney, 2001). Nevertheless, a few issues related to this technology still need to be solved, especially those related to the separation efficiency, filtration rate, and membrane fouling. Current findings in protein separation via membrane filtration can be adapted as an effort to successfully apply this technology for bromelain purification. Besides, the process optimization should be further investigated to maximize the extraction efficiency, minimize oxidation, avoid denaturalization, and increase protein acquisition (Arshad et al., 2014).

## 10.5 Membrane filtration process for bromelain extraction

In general, membrane filtration is a separation process of a particulate matter in a continuous liquid using semipermeable materials. It is a pressure-driven process, used for size-based separation that can be classified as either MF, UF, nanofiltration (NF), or reverse osmosis (RO), depending on the membrane pore size, as shown in Fig. 10.3. The membrane pore size is determined by the absolute size of the rejected particles (normally in μm). Besides, the membrane pore size can also be classified according to the molecular weight cutoff (MWCO). The MWCO is defined as the lowest solute molecular weight (MW) that can reach 90% membrane retention and is typically expressed in Dalton (Da) unit. For protein separation application, only MF and UF membranes are normally used to separate and purify proteins, including the bromelain enzyme.

The membrane filtration process can be performed either by dead-end or cross-flow filtration modes, depending on its application and scale. In a dead-end filtration mode, the feed passes through the membrane perpendicularly while the permeate is released at the other end. Particulate matter is trapped and accumulated on the filter as the filtration yield. This filtration mode is normally used at small laboratory scales which require the separation of macromolecules in a small volume (Cheryan, 1986). For some larger-scale applications cross-flow filtration is used, in which the separation happens when the feed is moving tangentially across the membrane surface.

**TABLE 10.6** Purification techniques for bromelain extraction (Arshad et al., 2014).

| Technique | Description | Approaches | Recovery (%) | Purity increment (fold) | Reference |
|---|---|---|---|---|---|
| Chromatography | Protein separation occurs when samples migrate through the column at different rates and conditions of solid stationary and mobile phases. This technique consists of two types of separation mechanisms: (1) adsorption, for example, ion exchange, affinity, hydrophobic interaction; (2) nonadsorption (e.g., gel filtration) | • Affinity chromatography | 94.8 | – | Chen et al. (2008) |
| | | • Immobilized metal affinity membrane | 94.6 | 15.4 | Nie et al. (2008) |
| | | • Expanded bed adsorption | – | 13 | Silveira et al. (2009) |
| | | • Ion-exchange chromatography | 84.5 | 10 | Devakate et al. (2009) |
| Precipitation | Protein separation happens in the cell extract with the addition of salts, polar solvents, nonpolar solvents, and organic polymers extracts by varying the temperature or pH. Protein solubility decreases with the increase of precipitating agent concentration, resulting in protein precipitation | • Ammonium sulfate precipitation | 82.1 | 2.5 | Hebbar et al. (2012) |
| | | • Ammonium sulfate precipitation | 69.7 | 3.0 | Devakate et al. (2009) |
| | | • Isoelectric point precipitation | – | 2.2 | Silvestre et al. (2012) |
| | | • Ethanol precipitation | 98 | 2.3 | Soares et al. (2012) |
| | | • Acetone precipitation | 86 | 4.9 | Chaurasiya and Hebbar (2013) |
| | | • Ethanol precipitation | – | 3.1 | Silvestre et al. (2012) |
| Aqueous two-phase system (ATPS) | Protein is separated into two immiscible phases after the mixing of polymer and salts or the combination of two incompatible polymers (PEG and dextran) in an aqueous solution. Factors that may influence the distribution of protein samples between these two phases include ionic strength, pH, temperature, molecular weight of the polymer and type of salt | • Polyethylene glycol (PEG)/($K_2HPO_4$/$KH_2PO_4$) aqueous two-phase system (ATPS) | 93.1 | 3.2 | Hebbar et al. (2012) |
| | | • PEG/$MgSO_4$ ATPS | 108.4–206 | 2.2–3.44 | Ketnawa et al. (2011), Ketnawa et al. (2010), Ketnawa et al. (2009) |
| | | • PEG/$K_2SO_4$ ATPS | 228 | 4.0 | Babu et al. (2008) |
| | | • Block copolymer ATPS | 79.5 | 1.3 | Rabelo et al. (2004) |
| | | • PEG/MgSO4 ATPS | 335.3 | 25.8 | Novaes et al. (2013) |
| Reverse micelle system (RMS) | Protein is separated as droplets from an organic solvent by the presence of surfactant (micelle). This happens due to the polar (hydrophilic) and nonpolar (hydrophobic) regions of the micelle, which induce the protein movement into the micelle. This movement is affected by pH, ionic strength of aqueous phase, and surfactant concentration | • Cetyltrimethylammonium bromide (CTAB)/isooctane/hexanol/butanol Reverse Micellar System (RMS) | 106 | 5.2 | Hebbar et al. (2008) |
| | | • CTAB/isooctane/hexanol/butanol RMS | 97.6 | 4.5 | Hemawathi et al. (2007) |
| | | • CTAB/isooctane/hexanol/butanol RMS | 78.9 | 4.0 | Chaurasiya and Hebbar (2013) |
| | | • Benzil Dodecyl Bis(hydroxyethyl) Ammonium Chloride (BDBAC)/ isooctane/hexanol RMS | – | 4.9 | Fileti et al. (2010) |
| | | • Affinity-based RMS | 228 | 12.3 | Kumar et al. (2011) |

| | | | | | |
|---|---|---|---|---|---|
| Membrane filtration | Protein fractionation and concentration are done according to size by using semipermeable membranes. It normally involves several filtration stages, where at the final filtration stage bromelain is retained in the system and further recycled until the desired concentration is achieved | • Microfiltration (MF) and ultracentrifugation | 85–100 | – | Lopes et al. (2009, 2012) |
| Integrated techniques | Different purification techniques are combined to get high bromelain purity and recovery | • MF, ammonium sulfate precipitation, ultrafiltration (UF), and ultracentrifugation | 50 | – | Doko et al. (1991) |
| | | • Nano-$TiO_2$ absorption and two-stage UF | 64.8 | 5.3 | Chao et al. (2009) |
| | | • High-speed countercurrent chromatography and reverse micelle system | 95.5 | 5.6 | Yin et al. (2011) |
| | | • Reverse micelle system and UF | 94.2 | 5.7 | Yin et al. (2011) |
| | | • Reverse micelle system and UF | 95.8 | 8.9 | Hebbar et al. (2012) |

**TABLE 10.7 Comparison of different purification strategies (Arshad et al., 2014).**

| Type of purification | Advantages | Disadvantages |
|---|---|---|
| Chromatography | • Highly specific and scalable | • Low separation efficiency and high cost<br>• Small sample loading capacity<br>• Multiple chromatography steps |
| Precipitation | • Upscalable with simple equipment requirement<br>• Low energy needed<br>• Effective for removal of trace contaminants | • Need dialysis step to remove excessive salt during precipitation |
| Aqueous two-phase system | • Low cost and shorter separation time<br>• Upscalable and produces higher yield<br>• Effective for removal of trace contaminants | • Difficult to recover target protein from phase-forming polymer |
| Reverse micelle system | • Upscalable and can be operated in a continuous mode<br>• No loss of native protein and activity<br>• Low interfacial tension | • Difficult to recover target protein from surfactant-containing organic solvent |
| Membrane filtration | • Concentrates the protein<br>• Effective for removal of trace contaminants | • Long separation process<br>• Increased protein loading will cause UF membrane clogging |

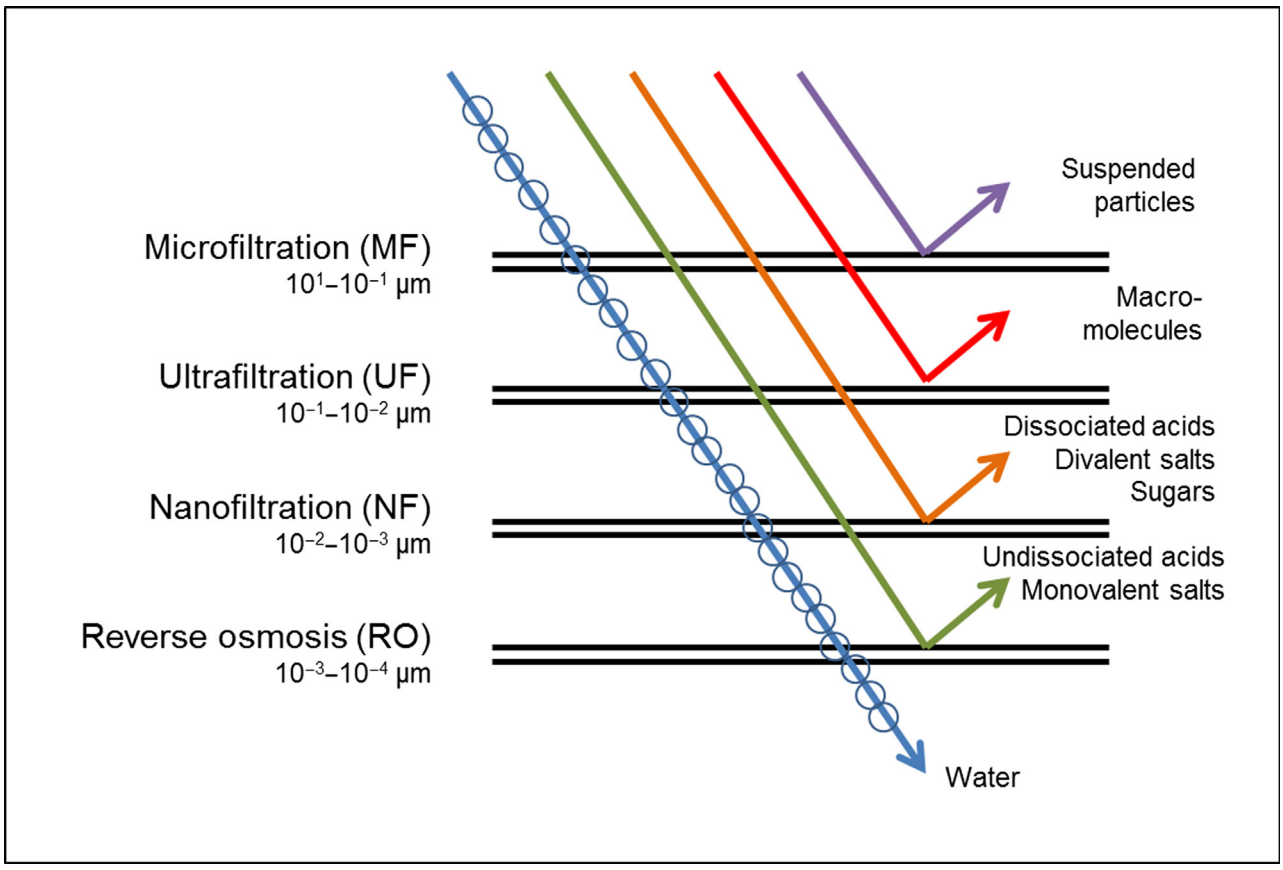



**FIGURE 10.3** Different types of membrane separation (Nor, 2017).

## 10.6 Application of membrane technology in bromelain purification

Numerous attempts have been focused on membrane-based processes for bromelain production either as a single process or in combination with other purification techniques. A multistage membrane process for the separation and concentration of bromelain from different pineapple parts has been described by Lopes et al. (2009), Gimeno et al. (2010), and Nor et al. (2016), which involved MF and UF. Several studies on the integration of membrane process with other purification technologies have been reported, such as with ammonium sulfate extraction (Doko et al., 1991), RMS (Hebbar et al., 2012), ATPS (Babu, 2008), and nano-$TiO_2$ absorption (Chao et al., 2009). Studies on utilizing membranes for the purification of bromelain are summarized in Table 10.8 (Nor et al., 2017). Despite this promising laboratory work, the potential to implement the outcomes to large-scale bromelain production still needs to be explored.

The membrane processing efficiency, particularly associated with the permeate flux and membrane fouling behavior, is strongly related to the composition of the feed. Extracts from plants including pineapple usually contain various components such as polysaccharides (soluble and insoluble fibers), proteins, pigments, and minerals. The composition is influenced by various factors including the type of plant, varietal characteristics, maturity, natural variation, climate, and cultural practices (Girard and Fukumoto, 2000). Very large components in the plant extract typically consist of soluble and insoluble complexes, colloids (mainly polysaccharide complexes), microorganisms, and starch. Proteins, including enzymes, and pectin are the medium-sized entities while small compounds in the plant extract typically include numerous sugars, organic acids, amino acids, phenolic compounds, pigments, essential oils, vitamins, and minerals (Fig. 10.4, Nor et al., 2017).

The targeted enzyme can be separated from the other components based on size exclusion. Very large particulates can easily be discarded by using centrifugation and MF process, whereas NF or small MWCO UF can be used to filter the small size components from the targeted enzyme.

Nevertheless, the separation of enzyme and protein from the pectin component is complicated and requires a precise separation based on their size. Bromelain enzyme has an MW of c.23.4−35.73 kDa (Arshad et al., 2014) while pectin has MW of 10−500 kDa with a global weight average of about 100 kDa (Girard and Fukumoto, 2000). Pectin is found to be responsible for the fouling build-up during membrane processing by forming high MW aggregates which in turn hinder membrane performance (Nor et al., 2015). It can be perceived as an impediment to the process, although it can also be considered as a valuable product for some types of juice fruit processing. In the case of pineapple it was reported that this fruit contains a small but a noticeable yield of pectin (Mohamed and Hasan, 1995; Normah and Ku Hasnah, 2000). In addition, a high hemicellulose content, including galactomannans, arabinogalactans, and galactoglucomannans, and a natural gum (a neutral polysaccharide containing 70% sugars, which are predominantly galactomannans) are present in pineapple juice which can also cause a quick reduction in the UF flux rate (Grassin and Fauquembergue, 1996). Also 0.06% (w/v) crude pectin was reported in the crude pineapple waste mixture extract (Nor et al., 2015). In order to reduce the potential of larger molecules causing fouling during the membrane processing, pretreatment of the feed should be considered. The treatment may consist of hydrolyzing soluble polysaccharides, which are responsible for the high viscosity of the juice, as well as liquefying the nonsoluble polysaccharides, such as nonsoluble pectins, cellulose, hemicellulose, and lignin from cell walls (Tochi et al., 2009)—this includes the enzymatic treatment of the feed with pectinase and cellulase enzymes.

## 10.7 Configurational considerations

It is important to select a suitable membrane configuration and features for higher resolution of bromelain separation from the pineapple extract. While the following section discusses the application of membrane processing for bromelain extraction, a multistage system (or systems) may be configured to allow for robust component separation that would be further processed downstream. For this reason a multistage membrane processing with or without the application of other purification techniques appears the most feasible approach. This approach predominantly involves the separation of bromelain from other macromolecules during the initial filtration stage (usually MF) and the concentration of the enzyme in the final filtration stage (mainly UF) (Fig. 10.2; Doko et al., 1991; Lopes et al., 2009; Gimeno et al., 2010). The enzyme can be further purified and concentrated using either RO or direct osmosis (DO) with diafiltration (Babu, 2008) as steps in the preparation of the concentrate for drying.

Multistage membrane systems have been successfully used for isolating a protein of interest for various applications. Some of the systems included use of high-performance tangential flow filtration, known as HPTFF and introduced by van Reis et al. (1997), which employed various strategies of exploiting differences in both size and charge to achieve high resolution of protein selectivity. By performing the HPTFF system, particularly in a two-stage UF process using

**TABLE 10.8** Summarized investigations on bromelain separation by membrane filtration process (Nor et al., 2017).

| Approach | Sample | Pretreatment | Membrane details | | | | Processing conditions | | | | | Purification fold (by membrane process itself) | Enzyme yield (%) | References |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Membrane system | Filter material | Pore size | Surface area ($m^2$) | pH | TMP (bar) | Temp (°C) | Flow rate (mL/min) | Volume reduction factor (VRF) | | | |
| Two-stage membrane filtration, starting by MF followed by UF | Pineapple pulp extract in phosphate buffer | – | MF—flat sheet | Polyvinyl fluorite | 0.1 μm | 0.0225 | 7.0 and 7.5 | 0.05–0.15 | Room temp. | – | – | – | 85 | Lopes et al. (2009, 2012) |
| | | | UF—centrifugal filter | – | 10 kDa | – | 7.0 and 7.5 | – | 4 | – | 10 | – | 100 | |
| Two-stage UF system | Pineapple waste extract | – | UF stage 1—tubular | Ceramic (zirconium oxide) | 75 kDa | 0.0055 | 4, 5, 5, 7 and 8.5 | 2 | 10–40 | 6900 | 15 | – | 96.8 | Nor et al. (2016) |
| | | | UF stage 2—tubular | Ceramic (zirconium oxide) | 10 kDa | 0.0055 | 4, 5, 5, 7 and 8.5 | 2 | 10–40 | 69 | 15 | 2.5 | – | |
| Purification by multiple processing steps involving MF, UF, ammonium sulfate extraction, ultracentrifugation, and freeze drying | Pineapple fruit extract | 12.5 ppm antifoaming agent and 200 ppm hemicellulase | MF—tubular | Ceramic (zirconium oxide) | 8 μm | 0.2 | 8.5 | – | 30 ± 2 | – | – | – | – | Doko et al. (1991) |
| | | | UF—tubular | Polysulfone | 10 kDa | 0.46 | 8.5 | 3 | 30 ± 2 | – | 7.4–17.1 | – | 97 | |
| Integration of the purification technique by coupling reverse micellar extraction (RME) process with UF | Pineapple core extract in sodium phosphate buffer | – | Tangential flow filteration (TFF) | Celluose acetate | 5 kDa | 0.005 | – | 1 | 25 ± 2 | 3 | 5 | 1.5 | 92.4 | Hebbar et al. (2012) |
| Three-stage membrane filtration involving 2 MF processes and an UF | Pineapple waste (peel and core) | – | MF | – | 1.2 and 0.2 μm | – | – | – | – | – | – | – | – | Gimeno et al. (2010) |
| | | | UF—centrifugal filter | Regenerated cellulose | 10 kDa | 0.00076 | – | – | – | – | – | – | – | |
| Purification and concentration process involving the aqueous two-phase system (ATPS), UF and direct osmosis (DO) | Pineapple core and peel extract in sodium phosphate buffer | – | UF—flat sheet | Polysulfone | 10 kDa | 0.0016 | 7 | 1–4 | 25 ± 2 | – | – | − 1.2 | – | Babu (2008) |
| | | | DO—flat sheet | Hydrophilic direct osmosis | – | 0.012 | – | – | 25 ± 2 | 100 | – | – | – | |
| Integration of the purification process by coupling nano-$TiO_2$ absorption with two-stage UF | Pineapple stem extract | – | – | – | 50 and 10 kDa | – | – | 2.5–3 | – | – | – | 1.4 | 64.7 | Chao et al. (2009) |

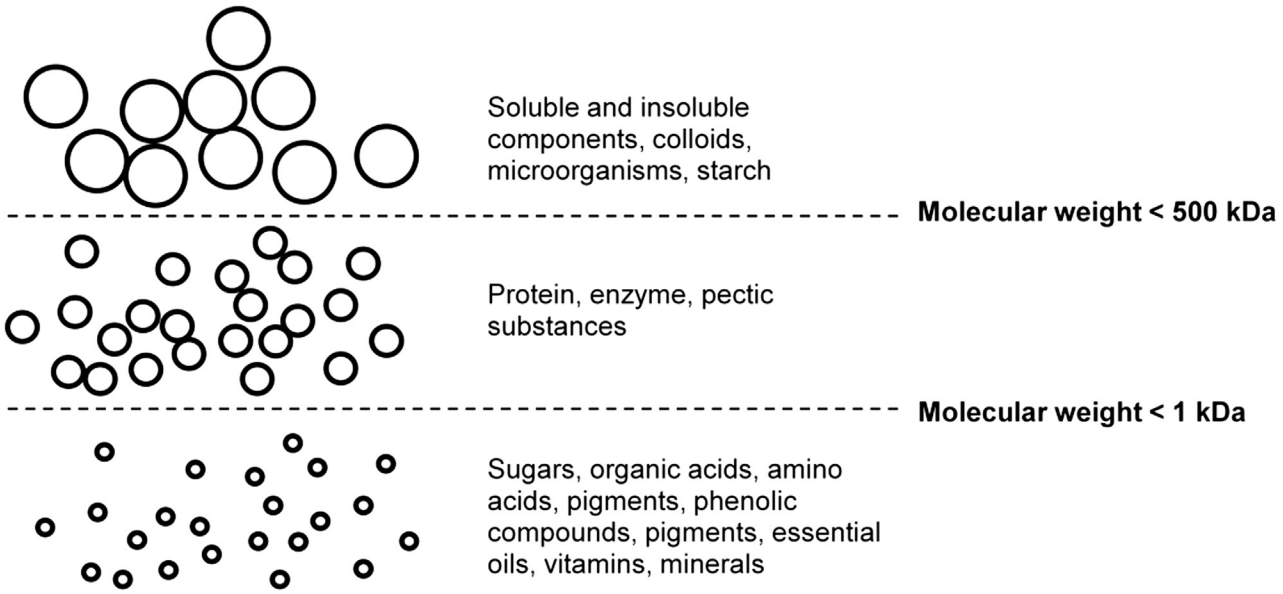



**FIGURE 10.4** Typical composition of plant extracts and corresponding MWs (Nor et al., 2017).

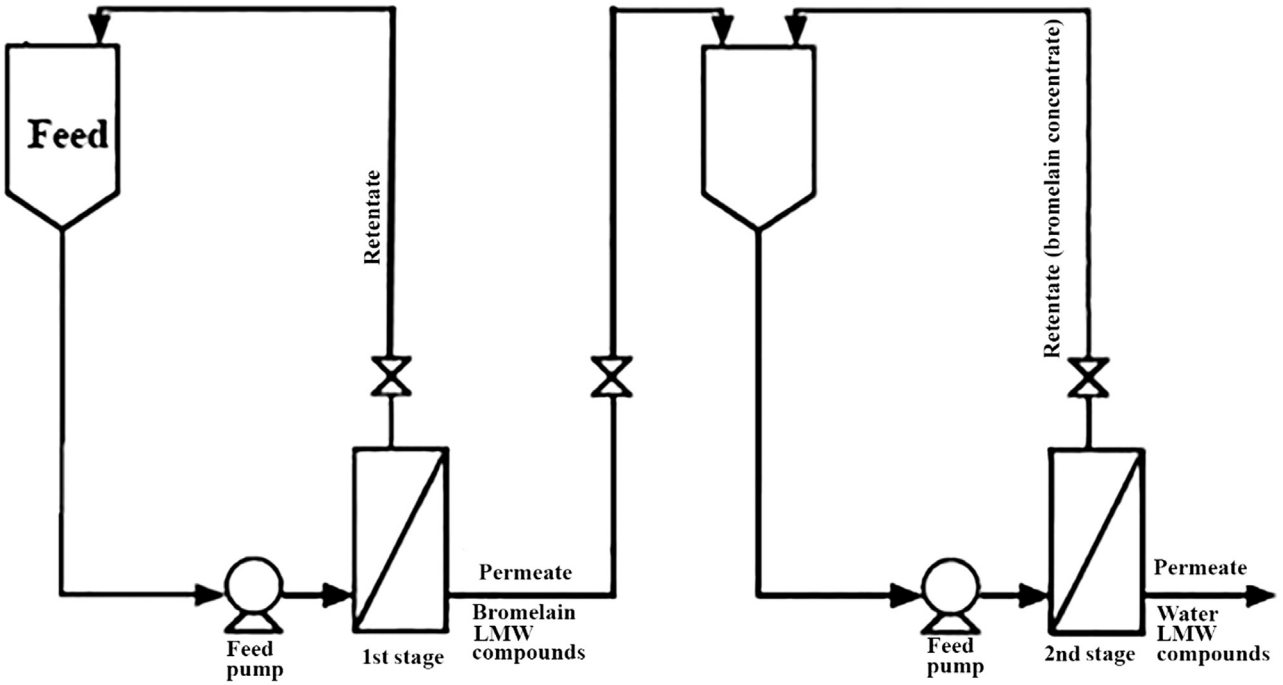



**FIGURE 10.5** A two-stage UF closed-loop cascade system for bromelain extraction (Nor et al., 2016).

membrane filters with MWCO of 150 and 10 kDa, they managed to separate bovine serum albumin (BSA) monomer from oligomers with purification factor of 9 and 86% yield. The two-stage UF strategy comprises two UF membrane filters with the pore size larger (stage 1) and smaller (stage 2) than the targeted protein of interest. The application of the two-stage UF strategy has also been reported for different types of proteins, such as the separation of α-lactalbumin and β-lactoglobulin from whey protein isolate (Cheang and Zydney, 2004), ovalbumin (OVA) from chicken egg white (Datta et al., 2009), and surfactin from fermentation broth (Isa et al., 2007), and can potentially be used for bromelain separation. For example, Nor et al. (2016) reported an increment of 2.5-fold purity of bromelain by applying the two-stage UF system which consisted of membranes with MWCO bigger (in the UF stage 1) and smaller (in the UF stage 2)

than the bromelain MW. Since bromelain has an MW ranging from 23.4 to 35.73 kDa (Arshad et al., 2014), for the two-stage UF application, the MWCO of the first stage membrane filter should be at least two times higher (75 kDa) and the second stage membrane filter two times lower (10 kDa) than that range (Nor et al., 2016) (Fig. 10.5).

The choice of membrane material in membrane processing is dependent on the processing pressure, temperature, and pH tolerance and chemical compatibility (Girard and Fukumoto, 2000). The membrane process for bromelain purification requires mild processing condition, usually between 0.5 and 4 bar, 10°C−30°C, pH 4 to 8.5 with no corrosive compounds or chemicals involved, which is suitable for many membrane materials. However, the cleaning procedure for the membrane is normally performed under more aggressive conditions with the application of strong acids or alkaline detergents at a higher temperature (60°C−80°C), which should be taken into consideration during membrane selection.

Most of the previous studies have used polymeric membranes, such as polysulfone, polyvinyl fluorite, and cellulose acetate, with the exception of Doko et al. (1991) and Nor et al. (2016) who used ceramic membranes (zirconium oxide). Ceramic membranes are classically more durable than the polymeric membranes. Examples of ceramic membranes include alumina ($\alpha$-$Al_2O_3$ and $\beta$-$Al_2O_3$), zirconia ($ZrO_2$), titania ($TiO_2$), glass ($SiO_2$), and silicon carbide (SiC) (Lee et al., 2015). They exhibit far superior mechanical, thermal, and chemical resistivity allowing much more extreme cleaning approaches without the risk of damaging the membranes (Lee et al., 2013). Furthermore, ceramic membranes have been reported to have a relatively narrow pore size distribution and higher porosity (Lee et al., 2002; Hofs et al., 2011), which are suitable for bromelain purification since better protein separation can be achieved using a membrane with a smaller pore size distribution (van Reis et al., 1997). Moreover, ceramic membranes exhibit lower organic fouling tendencies in certain circumstances compared with the polymeric membranes due to their hydrophilic and inorganic characters (Lee and Kim, 2014).

## 10.8 Processing parameters considerations

The feasibility of any membrane process can be affected by several processing parameters, including feed pH and concentration, transmembrane pressure (TMP), temperature, flow rate, and cross-flow velocity. Table 10.9 includes different processing parameters reported by various studies for the membrane process of bromelain. The feed pH appears to be one of the main processing factors (Lopes et al., 2009) as it influences the bromelain activity recovery by MF. The best recovery of activity appears to be obtained at pH 7.0 (Lopes et al., 2009). The same feed pH has been selected by Babu (2008) while Doko et al. (1991) chose to adjust the feed pH to 8.5 in their membrane-based process. To explore the role of pH more closely, Nor et al. (2016) investigated the effect of adjusting the feed pH on the flux behavior, enzyme recovery, and enzyme purity during the bromelain purification process. The pineapple extract used in their study was labeled as the crude waste mixture (CWM), which consisted of a specific ratio of different parts of pineapple waste including crown, peel, and core. The CWM extract was adjusted to four pH levels of 4, 5.5, 7, and 8.5. Improved flux behavior was achieved at feed pH 7, which might be strongly related to the viscosity reduction of the feed after the pH was changed away from its isoelectric point (pI) and thus improved the filtration rate in the membrane process. This is in agreement with their previous study (Nor et al., 2015), where observations on the effect of the CWM extract rheological properties at different pH also found a viscosity reduction when the pH was adjusted above its pI, since it leads to greater repulsion between the macromolecules.

The TMP is also a primary processing parameter which effectively increases the filtration rate. In general, higher TMP leads to a higher filtration rate, particularly in the membrane pressure-dependent region. However, the flux

**TABLE 10.9 Recommended improvements for membrane processing of bromelain separation (Nor et al., 2017).**

| Considerations | Recommended improvements |
|---|---|
| Various components in the feed | • Conducting enzyme pretreatment to minimize impact of other macromolecules during the membrane process |
| Efficiency of the process and purity of bromelain | • Implementing the HPTFF strategy particularly using the two-stage UF system, which consists of membranes with MWCO two times bigger and smaller than the molecular weight of bromelain<br>• Use of more robust membranes, that is, ceramic membranes<br>• Adopting other membrane processing approaches including diafiltration, gas sparging, etc.<br>• Optimizing relevant processing conditions including TMP, cross-flow velocity, pH, and temperature |
| Economy evaluation | • Establishing capital and investment cost. Operational cost should be further appraised based on the recommendations given above |

becomes independent of pressure due to the concentration polarization layer reaching its limiting concentration as the pressure increases further (Cheryan, 1986). Although high TMP would lead to a high filtration rate, the impact on enzyme activity should be considered (Lopes et al., 2009). The enzyme inactivation may take place due to conformational changes while passing through the membrane pores or due to contact with the fouling material. Table 10.8 shows the effects of high cross-flow rate and velocity parameters on bromelain separation. Flux normally increases with an improvement in hydrodynamic conditions on the surface of the membrane due to the high shear rates generated on the surface of the membrane, since these higher rates shear off more deposited material and consequently reduce the hydraulic resistance of the fouling layer (Cheryan, 1986). Selectivity of the membrane has been enhanced by increasing the cross-flow rate, since soluble protein and peptide transmission is promoted. For example, Datta et al. (2009) studied the effects of stirring speed on the separation of OVA from chicken egg white using a two-stage UF 50 and 30 kDa stirred cell. In both UF stages, the permeate flux was found to increase with the enhanced rate of stirring because of the enhanced turbulence at the membrane surface, resulting in a reduction of concentration polarization. Based on the data in Table 10.8, most of the studies have performed bromelain separation by membrane filtration at room temperature ($\sim$25°C−30°C). Increasing the processing temperatures may improve the permeation rate since at higher temperature, the membrane permeability coefficient and the diffusivity coefficient are higher, and the viscosity coefficient decreases (Girard and Fukumoto, 2000). This has been proven by Nor et al. (2016) who observed the effect of different processing temperatures, ranging from 10°C to 40°C, on the flux during the separation of bromelain by UF. The best flux was obtained at 40°C indicating the need to operate the process at the highest possible temperature for the maximum flux.

However, the stability of the enzyme during the operation also needs to be considered since high temperature may lead to the denaturation of the protein and the reduction of bromelain activity as well as potential fouling issues due to particle−particle interactions (Nor et al., 2015). Bromelain activity has been found to decline by 17% when the process was operated at 50°C for 60 min (Jutamongkon and Charoenrein, 2010), nevertheless the enzyme remained relatively stable at least for 1 week at room temperature (Hale et al., 2005). Furthermore, an increase of 3%−4% on the average processing capacity has been reported for every 1°C increase in operating temperature between 20°C and 60°C (Girard and Fukumoto, 2000), signifying the importance of ensuring the optimum balance between the membrane filtration rate, enzyme stability, and operating costs.

## 10.9 Bromelain purity

The purity of the separated bromelain after the membrane process is the main consideration in evaluating the performance of the membrane filtration. In general, purification of 1.2- to 2.5-fold have been reported (Table 10.8) by using the membrane process itself and the purity can be further improved by increasing the volume reduction factor (VRF) of the process (Nor et al., 2016). Besides increasing the VRF, it is believed that the bromelain purity can be further increased by implementing different filtration strategies such as using the HPTFF method as discussed in the previous section. By using this method, van Reis et al. (1999) reported 900-fold purification with a 90% yield when separating BSA from an antigen-binding fragment of monoclonal antibody (Fab). A similar work except using a complex multicomponent feed stream, that is, whey protein isolate, has demonstrated greater than 10-fold purification with 90% yield for the recovery of $\alpha$-lactalbumin (Cheang and Zydney, 2004). Moreover, for higher grade enzyme purity, the combination of the membrane technology with other purification methods can be considered. Performing the UF process after RMS in an integration purification process has resulted an increase of bromelain purity from 5.9-fold to 8.9-fold after the process (Hebbar et al., 2012). Purification of 1.5-fold can be achieved during the UF process itself after 5-fold volume reduction. In other studies the application of the membrane process with a combination of other purification methods has been found to exhibit varying levels of bromelain purity including an increase of 2.0 to 2.8-fold with the combination of ammonium sulfate extraction (Doko et al., 1991) and 5.29-fold when coupled with the nano-$TiO_2$ absorption process (Chao et al., 2009).

Nevertheless, most of the commercial applications of bromelain do not require a high purity of the enzyme except for the medical, pharmaceutical, and research areas. The enzyme is normally produced using less complex processes in high volumes for use in bulk production of food, feed, and fabrics (Illanes, 2008). Hence the bromelain purification process can be intuitively designed depending on the destination of the enzyme (Costa et al., 2014). The purification process can be employed to purify the protein of interest to the extent required for its final purpose (Jervis and Pierpoint, 1989). An increment of 2- to 4-fold purity has been recommended by Nor et al. (2015) for bromelain applications in the food industry, and this is achievable by using membrane filtration processes.

Economic consideration in the application of membrane-based processing for bromelain purification is crucial for the feasibility of the production. Lopes et al. (2012) provided an economic evaluation of bromelain production by

membrane processing. They analyzed the related bromelain production cost per hour and per day in Sao Paulo, Brazil, with the stipulation of the concentrated enzyme value of 125 mL/h and 1 L/day. The cost calculation included the cost of reagent, raw material, services, water, sewage, and energy involved, but the indirect costs, general materials, insurance, and depreciation were not taken into consideration. An estimated cost of US$12.68 to 16.03 per hour and US $99.27 to 128.23 per day has been reported to be 9−13 times lower than the same highly purified enzyme offered by pharmaceutical companies. In addition, the cost estimated in this study was 6.5−8.5 times less than the previous report of the economic analysis of bromelain production using liquid−liquid extraction technology. However, it is necessary to emphasize that the sale price of bromelain is not solely dependent on the production cost and might be influenced by other factors, such as the indirect overhead costs, demands, marketability of the product, etc., which may lead to a higher market price.

Overall the membrane technology presents a potential and attractive purification method for commercial bromelain production. Several recommendations for improvement have been discussed by adapting the successful studies on protein separation, and these are summarized in Table 10.9.

## 10.10 Valorization of carbohydrates

### 10.10.1 Insoluble fibers—cellulose and hemicellulose

As indicated in previous sections, pineapple waste contains very high amounts of cellulose and hemicellulose, which makes it a suitable candidate for the production of cellulose derivatives and composites with applications in food and polymer industries. The cellulose from pineapple peels may be derivatized into carboxymethyl cellulose and used in the preparation of environment-friendly packaging material (Chumee and Khemmakama, 2014). Cellulose composites or nanocellulose on the other hand have showed some promising applications in the biomedical field, due to their biocompatibility, low cytotoxicity, and biodegradability (Jorfi and Foster, 2015). The main characteristic of nanofibers is their large surface area, which allows the enhanced cell adhesion of proteins and pharmacological molecules while facilitating controlled drug-release (Leung and Ko, 2011). One of the major problems for the commercial application of these derived compounds resides in the complexity and feasibility of the extraction processes. For example, to prepare carboxymethyl cellulose, Chumee and Khemmakama (2014) obtained pure cellulose by refluxing pineapple peel powder with 0.5 M HCl and 1 M NaOH solution at 90°C for 1 and 2 h, respectively, followed by bleaching with calcium hypochlorite. The optimum condition for carboxymethylation was cellulose 5 g, chloroacetic acid 13.0 g, and 40% (w/v) NaOH 50 mL. Similarly, cellulose nanofibers are prepared after extraction to remove pectins and some hemicellulose by heating with 2% sodium hydroxide solution at 100°C for 4 h (Santos et al., 2013). After drying the dry extract is bleached with a solution made of equal parts acetate buffer and aqueous sodium chlorite at 80°C for 4 h. The bleached material is washed to achieve a neutral pH and then dried at 50°C for 12 h and milled. The powder is then acid hydrolyzed using sulfuric acid, centrifuged, and the precipitate is dialyzed for 4 days to obtain cellulose nanocrystals. These nanocrystals are needle-shaped, thermally stable, and highly crystalline with an average length of 249.7 nm and a diameter of 4.45 nm (Santos et al., 2013). Detailed methods for the extraction of cellulose nanofibers from various biomass materials have been extensively reviewed by Menon et al. (2017). Cellulose nanofibers are traditionally obtained by acid hydrolysis by partially dissolving cellulose fibrils (Feng et al., 2015). Naturally cellulose microfibrils are biosynthesized and self-assembled by repeatedly aggregating along cellulose chains creating crystalline and amorphous domains (Haafiz et al., 2014). These amorphous regions are preferentially hydrolyzed and removed when subjected to strong acid hydrolysis, while the crystalline regions are more acid resistant (Feng et al., 2015). Sulfuric acid is the most common hydrolyzing agent as it readily reacts with hydroxyl groups on the surface of cellulose through esterification. The process yields anionic sulfate ester groups which leads to the dispersion of the nanofibers in water by electrostatic repulsion due to enhanced electronegativity (Dufresne, 2013; Feng et al., 2015).

## 10.11 Soluble fibers—pectin and gums

Pectin presents a heterogeneous group of anionic structural polysaccharides found in fruit and vegetables and pectin is mainly prepared from waste citrus peel and apple pomace. It has a complex structure consisting of units of galacturonic acid as the main chain (Mohamadzadeh et al., 2010). α-L-Rhamnose units are occasionally inserted and the carboxyl groups are partially esterified by methyl alcohol. The extraction of pectin involves multiple steps, during which hydrolysis and extraction take place, and it is governed by a multitude of factors including temperature, pH, type of acid (inorganic or organic), and presence of chelating agents (EDTA, ammonium oxalate, or sodium hexametaphosphate)

(Kliemann et al., 2009). This type of pectin contains more than 50% of the carboxylic acid groups esterified, which makes it a high methyl ester pectin according to its gelling temperature. Modifying the extraction process or applying an acid treatment yields a low methyl ester pectin with less than 50% methyl ester groups. The pectin structure governs its application, and thus high methoxyl pectin gel is created via the formation of hydrogen bonds and hydrophobic interactions in the presence of acid (pH 3.0), which reduces electrostatic repulsions, and sugars, which modulate polymer−water interactions. Low methoxyl pectin on the other hand in the absence of particular cations would gel through cooperative associations at low temperatures (Constenla and Lozano, 2003). As the ability of pectin to form gel depends on the molecular size and degree of esterification, pectin from different sources does not have the same gelling ability. Therefore the industry is constant looking to identify new sources of this polysaccharide and establish its physicochemical properties.

Pineapple appears as a less suitable candidate for pectin extraction due to the fairly low concentration (approximately 15% of the total dietary fiber), and thus its extraction would be hindered by other major polysaccharides. Only a few reports are available in literature and all of them used some form of acid extraction due to the high yields achieved (Boobrad et al., 2006). In general the extraction process may involve use of highly concentrated organic or inorganic acids that usually solubilize pectin during heating at elevated temperatures for a certain period of time (Boobrad et al., 2006). Following solubilization pectin is precipitated either by ethanol or salt precipitation. Considering the conditions of extractions, any process design to extract pectin from pineapple waste needs to be deemed feasible. In many instances, such as the extraction of bromelain, pectin is rather considered a hindrance and additional steps may need to be incorporated into the process to allow for more efficient separation of bromelain from the remaining solids (Nor et al., 2017). Alternatively pineapple waste, in special reference to soluble fiber, could be used as a prebiotic. Prebiotics define a large group of food ingredients that are not digestible by human enzymes and thus transition into the gut where they are fermented by probiotics. Probiotic bacteria, mainly composed of bacterial genera of *Bifidobacteria* and *Lactobacilli*, once established in the human intestine have a beneficial effect on host health. Diaz-Vela et al. (2013) showed that dried pineapple peel waste had a positive effect on the growth of *Pediococcus pentosaceus* UAM22, *Aerococcus viridans* UAM21, and *Lactobacillus rhamnosus* GG indicating that the fibers were metabolized selectively by probiotics microorganisms. Similarly heat-treated pineapple biomass served as a promising prebiotic supplement in dairy and functional products, promoting the growth of probiotic microbiota and consequently the release of antimutagenic and antioxidant peptides (Sah et al., 2015).

## 10.12 Simple sugars—production of organic acids

Pineapple is rich in simple sugars mainly fructose, and thus the waste contains considerable amounts of fermentable sugars. Fermentation is typically used to convert these free sugars into different organic acids. The production of lactic, citric, succinic, and acetic acids from pineapple waste has been reported (Banerjee et al., 2017). Lactic acid is among the most important organic acids due to various applications in both food and biopolymer industries. Lactic acid was produced from pineapple syrup using a strain of *Lactobacillus lactis* (Ueno et al., 2003). On the other hand, Idris and Suzana (2006) used an immobilized strain of *Lactobacillus bulgaricus*, while fungal strains have also been used (Jin et al., 2005). The strains included *Rhizopus arrhizus* and *Rhizopus oryzae*, respectively. Citric acid is used as an acidulant to enhance the flavor of the products. It can be produced by solid-state fermentation of pineapple waste using *Aspergillus niger* (Kumar et al., 2003). The yield of citric acid can be improved with addition of *Yarrowia lipolytica* as the fermentation aid (Imandi et al., 2008). Succinic acid is another organic acid that is used as an acidity regulator. It also acts as a flavoring agent which adds sourness or astringency to the food. Pineapple waste serves as a suitable substrate for the production of succinic acid (Jusoha et al., 2014), yielding similar quantities of succinic acid to that obtained by the fermentation of glucose. Succinic acid can be derivatized into fumaric acid and dimethyl fumarate with various applications in the food and pharmaceutical industries. Acetic acid, or its more diluted commercial form, vinegar, can be produced from pineapple peel using a two-stage fermentation process (Sossou et al., 2009). Water is added to pineapple peel, heated, and then pressed to obtain juice which is then concentrated to 20% soluble solids and used for vinegar production through alcoholic and acetic fermentations. Acetic fermentation is carried out at 30°C using an acetic bacteria strain isolated from pineapple wine.

## 10.13 Other value-added products obtained from pineapple waste

A number of small MW solutes as well as possible derivatives from medium to high MW macromolecules can be obtained or derivatized from the pineapple waste. For example, 35 volatile compounds from pineapple rind and fibers

remaining after the juice extraction step were extracted and identified. The principal compounds were esters (37%), alcohols (29%), aldehydes (9%), ketones (9%), and acids (6%) (Barreto et al., 2013), which indicates that these extracts could be used as flavor-enhancing agents.

Another group of important commercial interests are polyphenols, which are secondary plant metabolites (Middleton et al., 2000). The major group of phenolic antioxidants includes phenolic acids, flavonoids, and tannins with a range of health benefits including antioxidant, antiallergenic, antimicrobial, antiinflammatory, and antithrombotic effects (Middleton et al., 2000). Quercetin, catechin, and kaempferol are among the widely studied polyphenolic compounds which are found to lower the risk of chronic diseases (Rasouli et al., 2017). Gallic acid, epicatechin, catechin, and ferulic acid have been identified as the main polyphenolic compounds in pineapple peel (Li et al., 2014). Due to their solubility in alcohols, the yield of polyphenols from pineapple peels is improved when methanol is used as the extracting solvent (Hossain and Rahma, 2011). The radical scavenging capacity of one mole of the polyphenolic compounds present in pineapple peel was found to be in the order: gallic acid > epicatechin = catechin > ferulic acid (Li et al., 2014).

Ferulic acid is another interesting compound with a potential application in the cosmetic and food industries (Ou and Kwork, 2004) since it can also be converted into vanillin (Lun et al., 2014). The process involves the bioconversion of ferulic acid to vanillic acid and vanillin. It is a three-step process (Lun et al., 2014). The first step involves the extraction of ferulic acid from pineapple waste powder that is then heat-treated, mixed with NaOH and $NaHSO_3$, shaken, and centrifuged. The pH of the supernatant is adjusted to 1.8 by adding HCl, and ferulic acid is extracted three times with ethyl acetate. The collected organic layer is concentrated and subject to solid-phase extraction. In the second step, the ferulic acid is transformed into vanillic acid by *A. niger* I-1472. Such produced vanillic acid is then biotransformed into vanillin by *Pycnoporus cinnabarinus* MUCL 39533. Vanillin is released into the supernatant after fermentation, which is then collected by adsorption onto an Amberlite XAD-4 resin.

Xylitol, as a sugar alcohol, has potential application as a low-calorie sweetener (Dasgupta et al., 2017). Chemically it is less reactive than xylose due to the absence of a reducing carbonyl group, which makes it difficult for tooth-decaying bacteria to utilize it in the production of energy and thus eliminates the generation of acid (Nayak et al., 2014). The wide application of xylitol, especially in the food industry, has been attributed to its low glycemic index (Khalid et al., 2012). Health concerns and greater awareness with regards to the importance of naturally derived low-calorie sweeteners have contributed to the increasing demand for xylitol. Pineapple waste, mainly the peel, is rich in xylan (~25%–35% dry basis), which is a potential substrate for the production of xylitol. Industrially xylitol is produced by catalytic hydrogenation of D-xylose to xylitol under high pressure and temperature (Dasgupta et al., 2017). The process has a great energy demand and involves highly specialized equipment in various and extensive purification steps for xylose and xylitol and the deactivation and recovery of a metal catalyst. For this reason, a more efficient process based on the bioconversion of lignocellulosic hydrolysates has been proposed, however this is still in an exploratory stage (Dasgupta et al., 2017).

Another compound of interest that can be derived from pineapple waste is xylooligosaccharides (XOS), which are the oligomers of β-1,4-linked xylose monomers, with various substituents including acetyl, phenolic, and uronic acids. XOS are available commercially, with a degree of polymerization (DP) ranging from 2 to 10. However, for food applications, XOS with a DP of 2–4 are preferred due to their maximum prebiotic effect (Vazquez et al., 2000). Their main advantage over a more established prebiotic, inulin, lies in their greater heat and acid stability which makes them good candidates for use in high-acidity systems such as juices (Vazquez et al., 2000). As a therapeutic XOS also possess antioxidant, antiallergenic, antimicrobial, and immunomodulatory activities (Moure et al., 2006). Due to purity requirements, the current process that could potentially be used for the conversion of hemicellulose from pineapple peel into XOS requires a highly efficient chromatographic method. Unfortunately this approach may not always provide financially viable yields (Choi et al., 2016), and thus further research is required in this area.

## 10.14 Conclusion

Pineapple waste offers a broad range of great opportunities to recover and produce high-value products. With this focus, particular attention in this overview has been given to some feasible approaches, especially the application of membrane processing, that would allow for the extraction and conversion of pineapple waste into high-value products. It is evident that even pineapple on-farm waste, including mainly pineapple leaves that are rich in cellulose and low in lignin, could be utilized for nanocellulose fiber recovery for specific applications, mainly in pharmacology and medicine for wound healing, drug delivery, and medicinal implants due to biocompatibility, low cytotoxicity, and biodegradability. Pineapple processing waste consisting of peel, pomace, core, and crown could be initially separated using membrane

technologies and then the range of streams could be further concentrated, such as in case of bromelain, or converted into xylitol, XOS, or biopolymers production, which have great potential applications in the food industry as processing aids, artificial sweeteners, prebiotic supplements, and food packaging material. These valorization approaches not only address the environmental issues but also create commercial opportunities. Certain challenges would persist such as year-round supply; however, these approaches may be used on multiple streams and waste sources.

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